Artificial Intelligence - MCQ - Slot C

Total points19/20

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Name of the Student

Preethi R

Registration Number

192511352

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CO2-AT1

An agricultural robot waters crops.

It measures soil moisture, weather forecasts, and water availability before deciding.

Which agent architecture best fits?

*

1/1

Simple Reflex Agent
Goal-Based Agent
Utility-Based Agent

Table-Driven Agent

A warehouse robot always moves toward the shelf that appears closest to the destination using straight-line distance but sometimes gets trapped behind obstacles.

Why does this happen?
*

1/1

Greedy Search ignores actual path cost.

BFS is incomplete.
DFS uses heuristics.
Uniform Cost Search uses heuristics.

A rescue robot searches every room of a collapsed building. Every movement costs the same.

If the nearest victim must be found first, which search strategy is appropriate?
*

1/1

DFS
BFS

Hill Climbing

Stimulated Annealing

An ambulance navigation system considers both:

Distance already travelled Estimated remaining distance

Which search algorithm naturally supports this?
*

1/1

BFS
DFS
Greedy Search
A*

Why is 'Stochastic Hill Climbing' often superior to standard 'Steepest-Ascent Hill Climbing' in complex
landscapes? *

1/1

It chooses randomly among uphill moves, where the probability of selection can vary with the steepness of the move, reducing susceptibility to local maxima traps.

It guarantees finding global optima in polynomial time.
It eliminates the need for neighbor generation functions.
It evaluates global state graphs using priority queue limits.

Why is 'IDS' (Iterative Deepening Search) often preferred for large search spaces where the goal depth is unknown, even if memory is not the main constraint?
*

1/1

It handles infinite paths better.
It reduces the branching factor.
It is faster than A*.
It combines BFS optimally and DFS space

In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5.
What is the value of f(n)?
*

1/1

12

2
35
7

An airline wants to assign thousands of flights to airport gates while minimizing passenger walking distance.

Which AI technique is most suitable?
*

1/1

BFS
DFS
Local Search

Bidirectional Search

A heuristic function sometimes overestimates the actual cost to reach the goal. This violates the admissibility condition required for optimal solutions in A* search.
What happens in such a case?
*

1/1

A* is guaranteed to find the optimal solution.
A* may find a solution faster, but it is not guaranteed to be optimal.

A* behaves exactly like Uniform Cost Search.
A* will fail to find any solution.

A thermostat monitors room temperature (input) and switches the heater on or off (output) accordingly, without maintaining any internal record of past states.
Which type of agent does this best represent? *

1/1

Goal-Based Agent
Utility-Based Agent
Simple Reflex Agent

Learning Agent

A drone is assigned to inspect wind turbines. During flight, strong winds change its position, birds suddenly appear, and GPS signals occasionally disappear.

Which environment classification is MOST appropriate?
*

1/1

Fully observable, deterministic, static
Partially observable, stochastic, dynamic

Deterministic, static, discrete
Fully observable, sequential

A self-driving car has access only to its sensor data and cannot know about a sudden pedestrian hidden behind a parked truck, yet it still makes the best possible decision given what it knows.

This illustrates that rationality depends on: *

1/1

Omniscience
The performance measure only

Percept sequence and available knowledge, not omniscience

Random action selection

You are building a search heuristic for an 8-puzzle. You have two heuristics: h_1 (number of misplaced tiles) and h_2 (total Manhattan distance).

Why is h_2 generally preferred over h_1 in A* search?

*

1/1

It is easier to compute.
It is more consistent.
It handles illegal moves.
It dominates h1

In a minimax tree, each node has 3 children and the depth of the tree is 3. The total number of leaf nodes depends on the branching factor raised to the depth.
How many leaf nodes are present?

*

1/1

9
12
16
27

A self-driving car must plan a route through a city. The environment is 'dynamic' because other cars move independently. What is the most significant challenge this poses for a 'Problem-Solving Agent' using classical search?
*

0/1

Heuristics become inadmissible.
Branching factor is infinite.
The solution is not executable
The state space is unknown

Correct answer

The solution is not executable

A researcher claims that a machine can be considered intelligent if a human evaluator cannot reliably distinguish its responses from those of another human during a text-based conversation.

Which foundational concept does this describe? *

1/1

Rational Agent Model
Turing Test

Chinese Room Argument
Expert System Approach

A self-driving car detects a pedestrian unexpectedly crossing the road.

What should a rational agent do?
*

1/1

Stop or safely avoid the pedestrian while minimizing overall risk.

Continue at the same speed.
Ignore the pedestrian.
Wait for another vehicle.

A company uses Hill Climbing to optimize delivery schedules. The algorithm repeatedly stops before finding the best schedule.

What is the MOST likely reason?
*

1/1

Local optimum

Infinite heuristic
Complete search tree
Breadth expansion

A search agent is trying to find the shortest path in a graph. It uses A* with $h(n) = 0$. This configuration is equivalent to:
*

1/1

Breadth-First Search.
Uniform-Cost Search.

Greedy Best-First Search.
Depth-First Search.

An autonomous delivery robot operates inside a hospital. It detects patients, avoids obstacles, changes its path when corridors are blocked, and delivers medicines on time.

Which characteristic best demonstrates that the robot behaves as a **rational agent**?
*

1/1

It always follows the shortest path.
It chooses actions that maximize successful delivery based on current information.

It memorizes every hospital corridor.
It never changes its route.

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